In some situations, an error won't cause an exception. Instead, middleware might generate an error status code. One such case occurs when a requested path isn't handled. In that situation, the pipeline returns a 404 error.

For APIs, which typically are consumed by apps (as opposed to end users), that result probably is fine. But for apps that typically generate HTML, such as Razor Pages apps, returning a 404 typically results in a generic, unfriendly page being shown to the user, as you saw in figure 4.8. Although this behavior is correct, it doesn't provide a great experience for users of these types of applications.

Error-handling middleware attempts to address these problems by modifying the response before the app returns it to the user. Typically, error-handling middleware returns either details on the error that occurred or a generic but friendly HTML page to the user. You'll learn how to handle this use case in chapter 13 when you learn about generating responses with Razor Pages.

The remainder of this section looks at the two main types of exception-handling middleware that's available for use in your application. Both are available as part of the base ASP.NET Core framework, so you don't need to reference any additional NuGet packages to use them.

4.3.1 Viewing exceptions in development: DeveloperExceptionPage

When you're developing an application, you typically want access to as much information as possible when an error occurs somewhere in your app. For that reason, Microsoft provides `DeveloperExceptionPageMiddleware`, which you can add to your middleware pipeline by using